

COMPUTATION AND VERIFICATION REPORT: R42

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This report specifies the new execution, exact proof objects, independent verification, preservation checks, and the provenance of the inherited finest nonlinear audit. It accompanies the article and its technical supplement and does not substitute software success for a mathematical assumption.

1. SCIENTIFIC AND REPOSITORY IDENTITY

The base is R41 commit 70aa68e2ecac000e16fb3a6b2c2f5ee275b6b64f, whose parent contains the R40 referee report. The addressed report is reviews/2026-09-25-econometrica-r40/referee_report.md. R42 uses the new branch revision/econometrica-r42-atomic-structure-2026-09-25. Its initial protocol commit is a6c6440b67210a4e6d1d75b372b5c4e59c6691a0. The protocol fixed cases before execution but did not freeze the final source. It is therefore not labeled a fully code-frozen holdout.

The original R40 source, all earlier PDFs and data, the report, and R41 files are preserved. The current index is the only inherited navigation file replaced; its previous content is copied into the R42 history directory. A publication check compares every inherited Git tree entry against the new branch, excluding that explicitly archived navigation file. It rejects deletion or modification of any other inherited object. The R42 sources, results, proof objects, reports, and historical volume are additive.

2. REPRODUCTION

Use Python 3.11 or later with NumPy and SciPy. The exact checker and repair tests use the Python standard library only. LaTeX compilation uses the unchanged repository `econsocart` class and configuration. The execution environment and actual dependency versions are retained in `results/environment.json`. From the repository root run:

```
export OPENBLAS_NUM_THREADS=1
export OMP_NUM_THREADS=1
python revisions/2026-09-25-r42/replication/run_suite.py
python revisions/2026-09-25-r42/replication/repair_tests.py
python revisions/2026-09-25-r42/replication/density_transfer.py
python revisions/2026-09-25-r42/replication/assemble.py
for x in ECTA SUPP RESPONSE COMPUTATION; do
  latexmk -pdf -halt-on-error "${x}_R42.tex"
done
```

The suite reuses an existing proof object only to independently check it; to regenerate all LP proposals remove only the twelve `case*.json.gz` files in the R42 proof directory. Do not remove older revision objects. The publication workflow starts with no R42 proof objects, so its proposals are fresh. The density transfer independently checks both adjusted finite solves. Timings and solver proposals can vary by platform. The required invariant is correctness of each exact certificate, not identical floating-point output or identical wall time.

3. NEW RESULTS AND VERIFICATION

The twelve declared structural cases all have positive implementation cost. Their independent checks reconstruct the model contract, policy probabilities, every restart loss, initial implemen-

tation cost, and residual-corrected lower bound. The largest program has $64 \cdot 64 = 4096$ policy simplexes and 16,384 variables. These are exact certificates for the original continuous-state randomized objective through the analytic aggregation theorem. They are not a restriction of the original optimizer to an assumed finite policy sieve.

The density experiment has three declared perturbation sizes. Two permit both adjusted finite solves; one fails the sufficient tightening condition and is recorded as such. The finite check does not independently machine-prove the analytic density norm. The density calculation and finite proof checks are kept separate in the coverage map.

Forty-eight additional exact property tests exercise the slack-ladder repair on action-dependent three-state, three-action kernels with horizons 1, 2, 4, and 6 and discount factors 1, 0.95, and 0.8. They check all-state feasibility, weak improvement of operating value, the policy-TV bound, and the implementation-cost bound. The tests complement the analytic proof; a finite test suite is not a proof of a general theorem.

4. INHERITED EVIDENCE AND THE FINEST NONLINEAR AUDIT

The retained R41 run is GitHub Actions run 36066906929, which completed successfully on its recorded source commit. Its artifact was retrieved before R42 publication. The result is copied unchanged into `results/inherited_r41_finest.json`. It independently reconstructs the complete $T = 32, N = 256, \varepsilon = 1/2$ object. The exact inequality count, runtime, proof hash, and verifier hash are in that file. R42 incorporates this existing successful audit; it does not claim to repeat its approximately 1,596-second arithmetic reconstruction locally.

The primary maintenance results are likewise inherited, not rerun. Their median widths and signed deterministic savings are recomputed directly from the rational endpoints. All original positive randomized intervals remain open. The tighter nonlinear rows still have no certified upper policy. Historical audit success, source identity, and publication hashes are never used to assert otherwise.

5. VERIFICATION BOUNDARY

The independent structural checker does not import the constructor, NumPy, SciPy, or an optimizer. It does share the mathematical specification and the standard rational-arithmetic runtime. Thus its independence is algorithmic and implementation-level rather than independence from every possible common software fault. The structural identities and infinite-dimensional reduction are written proofs, not machine-checked formal developments. The models are designed known-model economies, not empirically identified institutions. All such boundaries are visible in the current article rather than being hidden in a manifest.